



STANDARD OPERATING PROCEDURE

Solar Panel Barcode Scanning & Quality Control

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Project	Midway — Afton, Virginia
Application	Panel Scanner (web app)
Document	SOP-PS-001
Version	1.0

1. Purpose

This procedure defines how field crews record and quality-check solar module (panel) serial numbers for the **Midway** project in Afton, Virginia using the Panel Scanner web application. Following this SOP ensures every installed panel is captured against its physical location (section & row), that a photo of each barcode is retained, and that a minimum of **10% of all panels** are independently re-scanned for quality control.

2. Scope

Applies to all personnel of Sun Rise Construction and Development LLC who scan, log, review, or QC solar panels on the Midway site. It covers device setup, scanning, manual entry, review/correction, the 10% QC re-scan, and the automatic Google Sheets export.

3. Definitions

- **Project** — the job site (e.g., Midway). Each project gets its own Google Sheet automatically.
- **Section** — a block/area within the project. Each section becomes its own tab in the sheet.
- **Row** — a row of panels within a section. Each row has an expected panel count.
- **Panel** — a single module; one scanned barcode = one panel entry, numbered within its row.
- **Operator** — the person scanning; their name is attached to every entry (no login required).
- **QC** — Quality Control: a second, independent scan of a panel's barcode that must match the original.

4. System Overview

The Panel Scanner is a mobile-first web app (works on Android & iPhone). It requires no login — the operator simply enters their name. Scans are saved instantly to a central store, multiple people can work the same project at once, and every scan is mirrored to a Google Sheet in the company Drive. No app install is needed; it runs in the phone's web browser over HTTPS.

5. Getting Started

- 1 Open the scanner link in your phone browser (e.g., the company scan tool URL). Tip: add it to your home screen.
- 2 When prompted, enter **your name** and tap **Start Scanning**. Your name is tied to every panel you log.
- 3 Allow **camera access** when the browser asks — it is required for scanning.

6. Project, Section & Row Setup

Before scanning, make sure the project structure exists. On Midway, scanners select the existing project, section, and row. Supervisors create these as needed:

- 1 Tap the **Midway** project (or **+ New Project** to create it).
- 2 Open a **Section** (or **+ New Section**, e.g., “Section 1”). Each section is its own sheet tab.
- 3 Open a **Row** (or **+ New Row**) and set the **expected panel count** for that row.

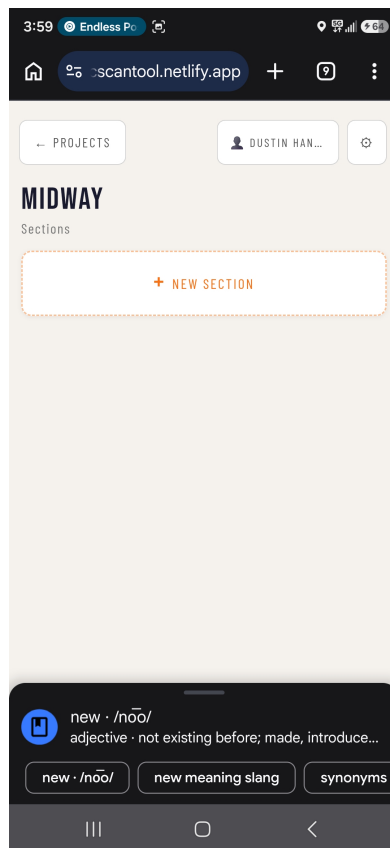


Figure 1 — Project view. Tap a section to open its rows; create new ones with the dashed tile.

7. Scanning Panels

With a row open, scanning is automatic and hands-free:

- 1 Tap **Start Scanning** to open the in-app viewfinder (a slim barcode band).
- 2 Center the panel's barcode in the band. It reads automatically — no shutter button.
- 3 On a successful read the screen flashes a green ✓ **COMPLETE**, beeps/vibrates, captures the barcode photo, and advances to the next panel number automatically.
- 4 Continue down the row. The panel counter increments for you.

Helpful behaviors

- **Steady capture** (on by default): briefly shows **HOLD STEADY...** to grab the sharpest photo. Hold the phone still for that moment.
- **Duplicate guard**: if a serial was already logged in the row, you get a **Duplicate Serial** warning — choose *Skip* or *Add Anyway*.
- **Over-target warning**: scanning beyond the row's expected count prompts **Row Already Full** — *Add Extra Panel*, *Finish Row*, or *Cancel*.
- **Completeness**: finishing a row prompts to start the next row; finishing all rows prompts a new section. Missing panel numbers are flagged before you move on.
- **Can't read a barcode?** Tap **Enter manually** to type the serial. (In photo mode, a failed read prompts *Retake Photo* or *Enter Manually* — the app never guesses a serial.)

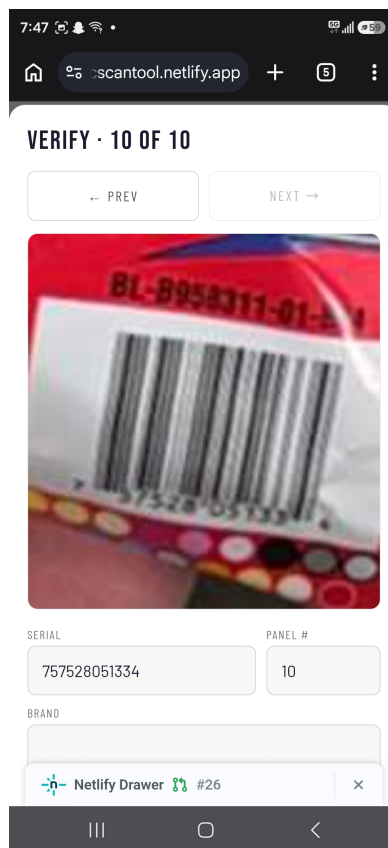


Figure 2 — A captured panel with its barcode photo and decoded serial.

8. Reviewing & Correcting Entries

Tap any logged panel to open the review view. Use ← **Prev** / **Next** → to step through the row, check each photo against its serial, edit the serial or panel number if needed, and tap **Save Edits**. Corrections update the central record and the Google Sheet (the matching row is replaced, not duplicated).

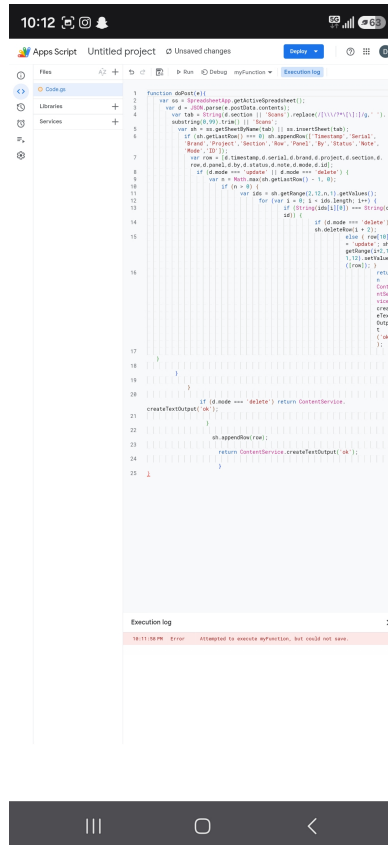


Figure 3 — Panel review view: photo, serial, panel number, and actions.

9. Quality Control — 10% Re-Scan

At least **10% of all logged panels** must be independently re-scanned. The project screen shows a **QC review** banner with progress toward the target (e.g., 1,770 of ~17,700 panels). To perform a QC check:

- 1 Open a panel in the review view (Section 8).
 - 2 Tap **QC Scan (scan again)** and re-scan the same physical barcode.
 - 3 If the re-scan **matches** the original serial, the panel is marked **QC PASS** and the app advances to the next panel.
 - 4 If it **does not match**, the panel is marked **QC FAIL** and shows what was scanned — investigate and correct the record.
- Each QC result writes to the Google Sheet: the second scan value, the Pass/Fail result, and (on Pass) an embedded image of the barcode.
 - Spread QC across sections/rows and operators so the 10% sample is representative.

10. Google Sheets Export

Every scan, correction, and QC result is sent to a Google Sheet in the company Drive. The app **auto-creates one spreadsheet per project** (in a “Panel Scanner” Drive folder), with **one tab per section**. Each row contains:

Section	Row	Panel	Serial	By
Project	Timestamp	QC Scan	QC Verified	QC Photo

One-time setup (supervisor / admin)

- 1 In the scanner, open ■ **Settings** and tap ■ **Copy script**.
- 2 Go to **script.google.com** → **New project**, delete the sample, paste the script, and Save.
- 3 **Deploy** → **New deployment** → **Web app**; set *Execute as: Me* and *Who has access: Anyone*; authorize Drive/Sheets access.
- 4 Copy the Web app URL (ends in **/exec**) and paste it into the scanner's Settings, then Save.
- 5 Scan a panel — the project's spreadsheet appears automatically in your Drive.



Figure 4 — Settings: paste the Apps Script Web App URL for Google Sheets sync.

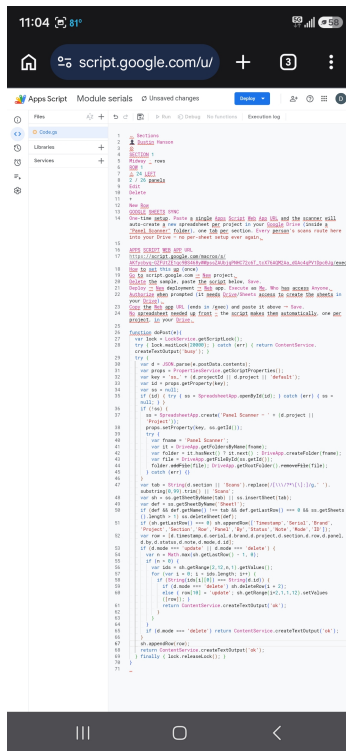


Figure 5 — Google Apps Script web app deployment.

11. Roles & Concurrent Use

- **Operators** scan panels; their name is recorded on every entry. Multiple operators can scan the same project simultaneously.
- **QC reviewers** perform the 10% re-scan checks and resolve any QC FAIL items.
- **Supervisor/Admin** maintains the project/section/row layout and the Google Sheets connection.

12. Troubleshooting

- **Camera won't open** — ensure browser camera permission is allowed and you're on HTTPS; reload the page.
- **Barcode won't read** — improve lighting, fill the band with the label, hold steady; or use **Enter** manually.
- **A button seems unresponsive after an update** — hard-refresh the page to load the latest version.
- **Sheet not updating** — confirm the Apps Script Web App URL is saved in Settings and deployed as “Anyone”.

13. Revision History

Version	Date	Author	Description
1.0	June 22, 2026	Dustin Hanson	Initial release.